F18CF – Linear Algebra

Tutorial Sheet 2 – Solutions

1. (a) We need to find $x, y \in \mathbb{R}$ such that

$$x \begin{pmatrix} 1 \\ -1 \end{pmatrix} + y \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} .$$

We reduce the associated system using Gaussian elimination:

$$\left(\begin{array}{cc|c} 1 & 1 & 5 \\ -1 & 1 & 3 \end{array}\right) \overset{\cdots}{\rightsquigarrow} \left(\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & 2 & 8 \end{array}\right)$$

By substitution, we get $y = 4$ and $x = 1$, hence:

$$\begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} + 4 \begin{pmatrix} 1 \\ 1 \end{pmatrix} .$$

(b) We have to find coefficients x, y, z such that

$$x \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + z \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix} .$$

We solve the associated system using Gaussian elimination. After performing the algorithm, we get

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ -1 & 0 & 1 & 3 \\ 0 & -1 & 1 & 1 \end{array}\right) \overset{\cdots}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 1 & 8 \\ 0 & 0 & 2 & 9 \end{array}\right)$$

By substitution, we get $z = 9/2$, then $y + z = 8 \rightsquigarrow y = 7/2$, and finally $x + y = 5 \rightsquigarrow x = 3/2$. Therefore:

$$\begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix} = \frac{3}{2} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \frac{7}{2} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \frac{9}{2} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} .$$

2. (a) We need to determine whether there exists $x, y \in \mathbb{R}$ such that

$$\begin{pmatrix} 5 \\ 3 \end{pmatrix} = x \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} + y \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} .$$

We reduce the associated system using Gaussian elimination:

$$\left(\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & 3 & 3 \end{array}\right) \overset{\cdots}{\rightsquigarrow} \left(\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -1 & 7 \end{array}\right)$$

By substitution, we get $y = -7$ and $x = 5$, hence:

$$\begin{pmatrix} 5 \\ 3 \end{pmatrix} = 12 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} - 7 \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} .$$

(b) We could again solve the system

$$a \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + b \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + c \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}$$

using Gaussian elimination. However there is a faster way here. Any linear combination of $(1, 2, 1)^T$, $(1, 3, 1)^T$ and $(1, 4, 1)^T$ will always have the same first and last component. That is, it will be of the form $(a, b, a)^T$ for some $a, b \in \mathbb{R}$. Therefore, the vector $(5, 3, 1)^T$ cannot be written as a linear combination of these vectors and does not lie in their span.

(c) We need to find values of a such that the equation

$$x \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + y \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ a \end{pmatrix}$$

has a solution. We simplify the associated system using Gaussian elimination:

$$\left(\begin{array}{cc|c} 1 & 1 & -2 \\ 2 & 1 & 1 \\ 3 & 3 & a \end{array}\right) \overset{\cdots}{\rightsquigarrow} \left(\begin{array}{cc|c} 1 & 1 & -2 \\ 0 & 1 & -5 \\ 0 & 0 & -a-6 \end{array}\right)$$

The system is consistent if and only there is no forbidden row. Thus, it has a solution if and only if $-a - 6 = 0$, i.e. $a = -6$. We then solve the system by substitution, which yields $y = -5$ and $x = 3$, hence

$$\begin{pmatrix} -2 \\ 1 \\ -6 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - 5 \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} .$$

3. (a) We have to find solutions of the system of linear equations in variables x, y, z:

$$x \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} + y \begin{pmatrix} 1 \\ 1 \\ 2 \\ -2 \end{pmatrix} + z \begin{pmatrix} 1 \\ 5 \\ 4 \\ -4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} .$$

We simplify the associated system using Gaussian elimination:

$$\left(\begin{array}{rrr|l} 1 & 1 & 1 & 0 \\ -1 & 1 & 5 & 0 \\ 1 & 2 & 4 & 0 \\ -1 & -2 & -4 & 0 \end{array}\right) \overset{\cdots}{\rightsquigarrow} \left(\begin{array}{rrr|l} 1 & 1 & 1 & 0 \\ 0 & 2 & 6 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The variable z is a free variable, so the system has infinitely many solutions, and in particular, the three vectors are not linearly independent. For instance for $z = 1$, we then get by substitution $2y + 6 = 0 \rightsquigarrow y = -3$ and $x + y + z = 0 \rightsquigarrow x = 2$, hence

$$2\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} - 3\begin{pmatrix} 1 \\ 1 \\ 2 \\ -2 \end{pmatrix} + \begin{pmatrix} 1 \\ 5 \\ 4 \\ -4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

(b) We have to find solutions of the system of linear equations in variables x, y, z:

$$x\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} + y\begin{pmatrix} 1 \\ 1 \\ 2 \\ 1 \end{pmatrix} + z\begin{pmatrix} 4 \\ 2 \\ 7 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

We simplify the associated system using Gaussian elimination:

$$\left(\begin{array}{rrr|l} 1 & 1 & 4 & 0 \\ -1 & 1 & 2 & 0 \\ 1 & 2 & 7 & 0 \\ -1 & -2 & -2 & 0 \end{array}\right) \overset{\cdots}{\rightsquigarrow} \left(\begin{array}{rrr|l} 1 & 1 & 4 & 0 \\ 0 & 2 & 6 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Thus, all the variables are pivot variables, so this homogeneous system has a unique solution: $x = y = z = 0$. In particular, the three vectors are linearly independent.

4. (a) No: A basis of $\mathbb{R}^3$ must have exactly 3 vectors.

   For (b) and (c), these 3 vectors form a basis if and only if their determinant (or volume) is non-zero. We compute the determinant using column expansion:

   (b) $\det\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & -1 \end{pmatrix} = \det\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} + \det\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = -1 + 1 = 0$, so these vectors do not form a basis of $\mathbb{R}^3$.

   (c) $\det\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & -1 \end{pmatrix} = \det\begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix} + \det\begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix} = -1 + (-3) = -4$, so these vectors do form a basis of $\mathbb{R}^3$.

5. (a) Let $a, b, c \in \mathbb{R}$ such that

$$a(\vec{u} + \vec{v}) + b(\vec{v} + \vec{w}) + c(\vec{w} + \vec{u}) = \vec{0}.$$

Rearranging this equation, we get

$$(a+c)\vec{u} + (a+b)\vec{v} + (b+c)\vec{w} = \vec{0}.$$

Since $\vec{u}, \vec{v}, \vec{w}$ are linearly independent, the coefficients must be zero, hence the system

$$\begin{cases} a + c &= 0 \\ a + b &= 0 \\ b + c &= 0 \end{cases}$$

which we solve by Gaussian elimination:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right) \overset{\cdots}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right)$$

All the variables are pivot variables, hence this system has the unique solution $a = b = c = 0$. We thus get $a = b = c = 0$, hence $\vec{u} + \vec{v}, \vec{v} + \vec{w}, \vec{w} + \vec{u}$ are linearly independent.

(b) We could as before solve the system

$$a(\vec{u} - \vec{v}) + b(\vec{v} - \vec{w}) + c(\vec{w} - \vec{u}) = \vec{0}$$

with Gaussian elimination, but here we immediately spot a relation, namely

$$(\vec{u} - \vec{v}) + (\vec{v} - \vec{w}) + (\vec{w} - \vec{u}) = \vec{0}.$$

Thus, the vectors $\vec{u} - \vec{v}, \vec{v} - \vec{w}, \vec{w} - \vec{u}$ are not linearly independent.

6. (a) We have $1 + 0 = 1$, hence $\vec{u} \in H$, and $0 + 1 = 1$, hence $\vec{v} \in H$.

(b) A vector of $\text{span}(\vec{u}, \vec{v})$ is of the form

$$a\vec{u} + b\vec{v} = a \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ b \\ a + b \end{pmatrix}$$

for some $a, b \in \mathbb{R}$, and we check that we indeed have $a + b = a + b$, hence $a\vec{u} + b\vec{v} \in H$. It follows that $\text{span}(\vec{u}, \vec{v}) \subseteq H$.

(c) Let us show the reverse inclusion. Let $\vec{w} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ be a vector of H. Since $\vec{w} \in H$, we have $z = x + y$, hence

$$\vec{w} = \begin{pmatrix} x \\ y \\ x + y \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = x\vec{u} + y\vec{v} \in \text{span}(\vec{u}, \vec{v}).$$

We thus have $H \subseteq \text{span}(\vec{u}, \vec{v})$, hence $H = \text{span}(\vec{u}, \vec{v})$

7. (a) Let $\lambda_1, \ldots, \lambda_k, \lambda \in \mathbb{R}$ such that

$$\lambda_1 \vec{v}_1 + \cdots + \lambda_k \vec{v}_k + \lambda \vec{v} = \vec{0}.$$

We need to show that all the coefficients are zero. First notice that we must have $\lambda = 0$. Indeed, if that were not the case, then we would have

$$\vec{v} = -\frac{\lambda_1}{\lambda}\vec{v}_1 - \cdots - \frac{\lambda_k}{\lambda}\vec{v}_k \in \operatorname{span}(\vec{v}_1, \ldots, \vec{v}_k),$$

which is not the case by assumption. Thus, $\lambda = 0$, and the equation becomes

$$\lambda_1 \vec{v}_1 + \cdots + \lambda_k \vec{v}_k = \vec{0}.$$

But since $\vec{v}_1, \ldots, \vec{v}_k$ is linearly independent, we get $\lambda_1 = \cdots = \lambda_k = 0$. We thus have

$$\lambda = \lambda_1 = \cdots = \lambda_k = 0,$$

hence $\vec{v}_1, \ldots, \vec{v}_k, \vec{v}$ is a linearly independent family.

(b) Since a family of strictly fewer than n vectors cannot be spanning by a result of the course, we can apply the procedure (a) until we get a linearly independent family $\vec{v}_1, \ldots, \vec{v}_n$ consisting of exactly n vectors. By a result of the course, such a family is automatically a basis of $\mathbb{R}^n$.

8. (a) We clearly have $\operatorname{span}(\vec{v}_1, \ldots, \vec{v}_{k-1}) \subseteq \operatorname{span}(\vec{v}_1, \ldots, \vec{v}_k)$. Let us prove the reverse inclusion. Let $\vec{w} \in \operatorname{span}(\vec{v}_1, \ldots, \vec{v}_k)$ and let $\lambda_1, \ldots, \lambda_k \in \mathbb{R}^n$ such that

$$\vec{w} = \lambda_1 \vec{v}_1 + \cdots + \lambda_{k-1}\vec{v}_{k-1} + \lambda_k \vec{v}_k.$$

Since $\vec{v}_k \in \operatorname{span}(\vec{v}_1, \ldots, \vec{v}_{k-1})$, let $\mu_1, \ldots, \mu_{k-1} \in \mathbb{R}$ such that

$$\vec{v}_k = \mu_1 \vec{v}_1 + \cdots + \mu_{k-1}\vec{v}_{k-1}.$$

We thus get

$$\begin{aligned}
\vec{w} &= \lambda_1 \vec{v}_1 + \cdots + \lambda_{k-1}\vec{v}_{k-1} + \lambda_k(\mu_1 \vec{v}_1 + \cdots + \mu_{k-1}\vec{v}_{k-1}) \\
&= (\lambda_1 + \lambda_k \mu_1)\vec{v}_1 + \cdots + (\lambda_{k-1} + \lambda_k \mu_{k-1})\vec{v}_{k-1} \\
&\in \operatorname{span}(\vec{v}_1, \ldots, \vec{v}_{k-1}).
\end{aligned}$$

This proves that $\operatorname{span}(\vec{v}_1, \ldots, \vec{v}_k) \subseteq \operatorname{span}(\vec{v}_1, \ldots, \vec{v}_{k-1})$, hence the equality.

(b) Since a family of strictly more than n vectors of $\mathbb{R}^n$ cannot be linearly independent by a result of the course, and being linearly dependent precisely means that one of the vectors lives in the span of the others, we can repeat the procedure in (a) sufficiently many times, removing one vector at a time, until we get a spanning sub-family $\vec{v}_1, \ldots, \vec{v}_n$ consisting of exactly n vectors. By a result of the course, such a family is automatically a basis of $\mathbb{R}^n$.